Network Traffic Analysis & C2/Exfiltration Detection for SOC

Week 1 Documentation: Network Foundations & Traffic Capture

Project	Project 1 – Network Traffic Analysis & C2/Exfiltration Detection for SOC
Week	1 – Network Foundations & Traffic Capture
Date	2026-10-31

1. Overview

The goal of this project is to build a SOC-oriented network traffic analysis capability, starting from understanding the normal network baseline and progressing to detecting C2 activity and data exfiltration.

In Week 1 we focused on networking fundamentals (OSI/TCP-IP, IP addressing, common protocols) and on capturing and analyzing traffic. The first task completed is the SOC Network Baseline Dashboard: a dashboard that summarizes normal traffic in the lab network and highlights unusual activity that deserves review.

Week 1 Objectives

- Review OSI/TCP-IP, IP addressing, subnetting, and common protocols (HTTP, DNS, SMTP, SMB, SSH, RDP).
- Set up Wireshark and use capture filters and display filters.
- Capture traffic: browsing, DNS, TCP handshake, and file transfer.
- Prepare the weekly deliverables.

Week 1 Deliverables

Deliverable	Status
Baseline traffic summary	✓ Done (Section 3)
Network protocol map	✓ Done (Sections 3.2 and 4)
Annotated PCAP	(To be attached / in progress)
Wireshark filter cheat sheet	(To be attached / in progress)

2. Lab Environment

- **Network:** Internal network 192.168.1.0/24; no internet egress was observed.

- **Capture files (4):** W-11.pcapng, W-Server.pcapng, kali_linux.pcapng, Ubunto_server.pcap
- **Tools:** Wireshark for capturing and analyzing traffic, and the Baseline Dashboard for summarizing results.

3. Task 1: SOC Network Baseline Dashboard

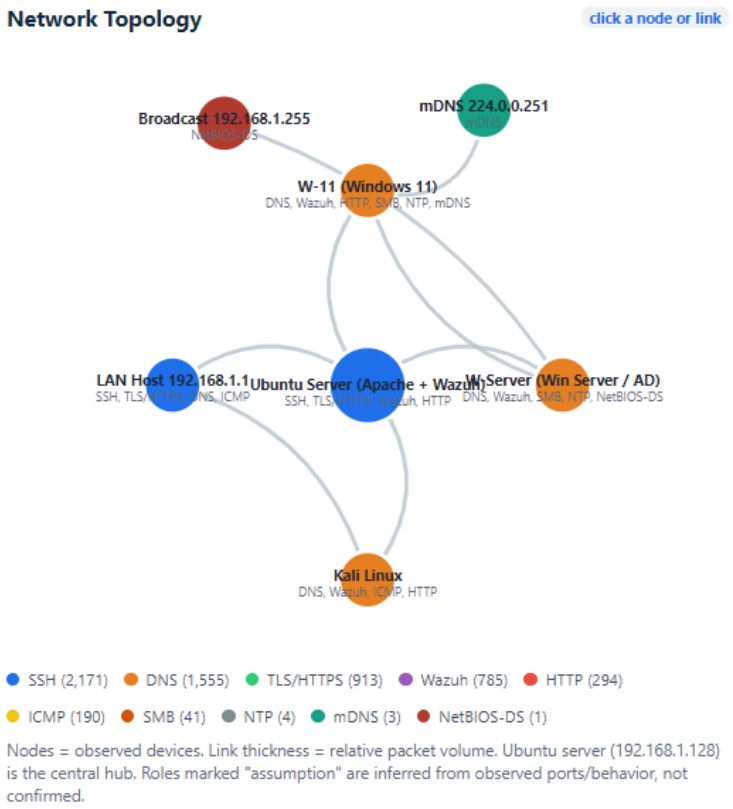
3.1 Dashboard Summary



Metric	Value
Devices observed (unique IPs)	7
Total packets (across 4 captures)	6,069
Traffic volume	1,578,420 bytes (~1.51 MB)
Distinct IP protocols	10
Findings to review	4

Summary: All traffic is internal within 192.168.1.0/24, and no internet egress appears in the captures. This represents the baseline for comparison in the coming weeks.

3.2 Network Topology

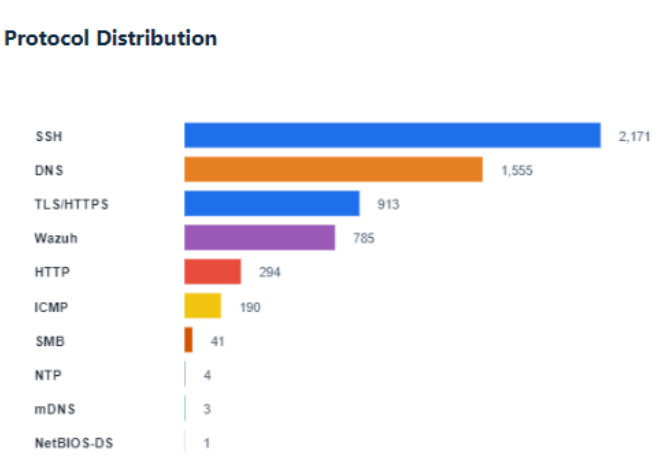


The diagram shows the devices observed in the captures and the relationships between them. Line thickness represents relative packet volume.

Device	Observed Protocols
Ubuntu Server (Apache + Wazuh) – 192.168.1.128	SSH, TLS/HTTPS, Wazuh, HTTP
W-11 (Windows 11)	DNS, Wazuh, HTTP, SMB, NTP, mDNS
W-Server (Windows Server / AD)	DNS, Wazuh, SMB, NTP, NetBIOS-DS
Kali Linux	DNS, Wazuh, ICMP, HTTP
LAN Host 192.168.1.1	SSH, TLS/HTTPS, DNS, ICMP
Broadcast 192.168.1.255	NetBIOS-DS
mDNS 224.0.0.251	mDNS

- The Ubuntu Server (192.168.1.128) is the central hub of the network.
- Device roles are inferred from observed ports and behavior and are not confirmed.

3.3 Protocol Distribution



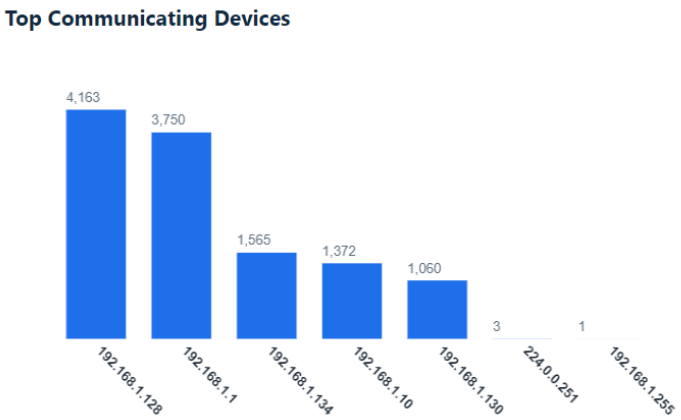
Protocol	Packets
SSH	2,171
DNS	1,555
TLS/HTTPS	913
Wazuh	785
HTTP	294
ICMP	190
SMB	41
NTP	4
mDNS	3
NetBIOS-DS	1

Analysis:

- SSH is the largest protocol by packet count, followed by DNS and then TLS/HTTPS.
- Wazuh traffic (785 packets) is expected, as the agents communicate with the manager.

- HTTP (294 packets) means unencrypted traffic exists on the network, which relates to finding UN-01.

3.4 Top Communicating Devices



IP Address	Packets
192.168.1.128	4,163
192.168.1.1	3,750
192.168.1.134	1,565
192.168.1.10	1,372
192.168.1.130	1,060
224.0.0.251	3
192.168.1.255	1

192.168.1.128 and 192.168.1.1 are the most active devices, which is consistent with the Ubuntu Server acting as the central hub.

4. Network Protocol Map

Summary of how protocols relate to devices (from the topology and protocol distribution):

Protocol	Role in the Lab	Seen On
SSH	Remote administration	Ubuntu Server, LAN Host
DNS	Name resolution	W-11, W-Server, Kali, LAN Host
TLS/HTTPS	Encrypted web traffic	Ubuntu Server, LAN Host
HTTP	Unencrypted web traffic	Ubuntu Server, W-11, Kali
Wazuh	Security monitoring (agent ↔ manager)	Ubuntu Server, W-11, W-Server, Kali
SMB	File sharing	W-11, W-Server
ICMP	Connectivity / ping	Kali, LAN Host
NTP	Time sync	W-11, W-Server
mDNS	Local name resolution (multicast)	W-11 → 224.0.0.251
NetBIOS-DS	Legacy Windows broadcast	W-Server → 192.168.1.255

5. Unusual Activity



None of these activities is confirmed malicious. They are observed patterns that warrant investigation.

ID	Severity	Finding	Evidence	Suggested Action
UN-01	Medium (Operational)	Cleartext authentication: HTTP POST to /login.php over port 80 without TLS.	4 login POST requests to Apache 192.168.1.128 from 192.168.1.130 and 192.168.1.134	Enforce HTTPS for login pages and review application authentication logs
UN-02	Low	DNS resolution failure: DNS responses with RCODE 2 (SERVFAIL).	505 SERVFAIL responses; top names: functional.events.data.microsoft.com (162), edge.microsoft.com (154), www.bing.com (82)	Check DNS server configuration (192.168.1.10) and zone records

ID	Severity	Finding	Evidence	Suggested Action
UN-03	Low	Repeated DNS queries (beacon-like pattern): the same domains are queried repeatedly. Consistent with OS telemetry but also resembles beaconing. No confirmed C2 and no follow-up TCP/TLS to the resolved IPs.	functional.events.data.microsoft.com ×189, edge.microsoft.com ×152, www.bing.com ×98, 2.debian.pool.ntp.org ×56, 3.debian.pool.ntp.org ×55, 0.debian.pool.ntp.org ×55	Correlate with endpoint process telemetry and confirm these are Windows telemetry
UN-04	Informational	mDNS / .local name queries: queries for .local (link-local) names.	318 queries, mostly *.localdomain names such as firefox.settings.mozilla.com.localdomain	Expected on flat LANs; no action unless multicast DNS is undesired

Related captures: UN-01 (Ubunto_server, W-11, kali_linux) · UN-02 (W-11, W-Server) · UN-03 (W-11, W-Server, kali_linux) · UN-04 (kali_linux)

6. Notes & Issues

- The Ubuntu server capture file is named Ubunto_server.pcap (original file name, spelling kept as-is).
- 192.168.1.130 and 192.168.1.134 appeared as the sources of the login requests; each IP still needs to be mapped to its actual device (W-11 / Kali) during manual review.
- UN-03 needs further verification (correlation with process telemetry) before it can be classified as normal telemetry.

7. Action Plan

Week	Focus	Key Tasks	Deliverables
1 ✓	Network Foundations & Traffic Capture	Review OSI/TCP-IP, subnetting, and protocols; set up Wireshark and filters; capture browsing, DNS, TCP handshake, and file transfer traffic	Network protocol map, Wireshark filter cheat sheet, annotated PCAP, baseline traffic summary
2	Protocol Analysis & Anomaly Identification	Analyze HTTP/DNS/SMB/TLS metadata and TCP streams; identify unusual ports, beaconing, DNS tunneling, and large outbound transfers; use Zeek/Suricata	Protocol analysis report, suspicious traffic findings, Zeek/Suricata alert review, IOC list, PCAP evidence
3	Recon, C2 & Exfiltration Detection	Detect Nmap scanning, OS fingerprinting, and service enumeration; analyze malicious PCAPs (C2 beaconing, DGA, exfiltration, web shell); MITRE ATT&CK mapping	Scan detection report, C2/exfiltration analysis, MITRE ATT&CK mapping, detection logic
4	AI-Assisted Network Triage & Reporting	Use Ollama/local LLM to summarize findings and draft IOC summaries; validate AI output manually; write the final SOC-style report	Final Network Threat Detection Report, AI prompt library, human-validated findings, executive summary, remediation/detection recommendations

Task Status

Task	Deadline	Status
Baseline Dashboard	Week 1	✓ Done
Wireshark filter cheat sheet	Week 1	Pending
Annotated PCAP	Week 1	Pending
Protocol analysis (Zeek/Suricata)	Week 2	Pending
Recon / C2 / Exfiltration detection	Week 3	Pending
AI-assisted triage & final report	Week 4	Pending